

DEWAR LAKES, NU, Canada

WMO: 710920

Lat: 68.651N

Lon: 71.234W

Elev: 1728

StdP: 13.80

Time Zone: -5.00 (NAE)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-39.1	-35.9	-41.2	0.5	-34.0	-40.0	0.6	-33.6	46.4	0.3	41.0	-6.9	8.4	290	1.150

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	58.8	48.4	55.1	46.5	51.8	44.9	48.8	56.2	46.9	52.9	45.1	50.1	10.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
44.8	46.6	48.8	42.9	43.4	48.5	41.1	40.4	46.9	20.2	56.2	19.1	53.0	18.1	50.1	54.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 37.6	32.4	28.5	-41.4	62.0	4.0	5.1	-44.2	65.6	-46.6	68.6	-48.8	71.5	-51.8	75.2		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	-16.4	-17.8	-12.9	-0.5	16.8	32.8	44.4	40.6	28.3	14.9	1.7	-8.8		
	DBStd	23.48	11.41	11.45	12.17	10.32	8.50	6.38	6.50	6.06	5.88	8.27	10.52	11.82		
	HDD50	14474	2058	1898	1949	1516	1031	516	200	298	651	1087	1448	1822		
	HDD65	19917	2523	2318	2414	1966	1496	966	640	756	1101	1552	1898	2287		
	CDD50	32	0	0	0	0	0	0	25	7	0	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	12.6	11.9	11.9	12.1	12.8	13.4	13.1	12.4	12.6	12.6	12.8	13.6	12.6		
	PrecAvg	9.8	0.2	0.1	0.2	0.4	0.8	0.7	1.8	2.5	1.3	1.0	0.5	0.2		
	PrecMax	15.9	1.2	0.6	1.9	1.3	2.0	2.7	4.1	6.0	2.9	2.7	1.7	1.1		
(15) Precipitation	PrecMin	4.2	0.0	0.0	0.0	0.0	0.1	0.0	0.1	0.2	0.3	0.4	0.0	0.0		
	PrecStd	3.4	0.3	0.2	0.4	0.3	0.5	0.7	1.1	1.7	0.7	0.6	0.4	0.2		
	0.4%	DB	22.6	18.0	21.8	28.4	36.9	53.3	65.5	60.3	43.8	32.5	27.0	24.2		
	MCWB	22.6	17.9	21.8	23.5	34.0	44.7	51.4	49.5	40.1	32.5	27.0	18.5			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2%	DB	12.2	9.6	15.2	23.1	33.4	48.7	61.4	56.0	40.2	29.8	23.3	18.5		
	MCWB	11.8	9.0	14.3	20.2	31.4	41.9	49.5	46.9	37.6	29.7	22.9	16.8			
	5%	DB	4.3	3.9	9.6	18.1	30.6	45.0	57.7	52.8	37.9	28.0	20.4	12.1		
	MCWB	3.3	3.2	8.4	16.9	29.4	39.5	47.9	45.8	36.1	27.8	19.9	11.2			
	10%	DB	-0.7	-1.9	4.4	13.7	28.0	42.0	54.5	49.9	35.8	25.7	16.9	6.5		
	MCWB	-1.5	-2.6	3.6	12.8	27.1	37.9	46.4	44.0	34.8	25.4	16.6	5.7			
	0.4%	WB	3.6	14.9	15.0	23.2	38.9	45.9	52.3	49.5	41.1	31.3	24.5	16.7		
	MCDB	4.1	15.1	15.6	26.8	44.2	52.0	63.1	56.3	43.1	31.3	24.5	18.4			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	1.1	9.6	8.7	17.4	33.3	43.3	50.2	47.5	37.8	29.3	21.0	11.4		
	MCDB	1.8	10.1	9.6	18.3	35.2	49.1	58.9	53.2	39.3	29.3	21.3	12.1			
	5%	WB	-2.5	3.7	3.6	14.0	31.3	41.0	48.3	46.0	36.0	27.6	18.0	7.0		
	MCDB	-1.9	4.2	4.3	14.7	32.7	46.5	55.8	51.1	37.1	27.7	18.4	7.7			
	10%	WB	-5.5	-2.0	-1.1	9.7	28.9	38.8	46.5	43.7	34.5	25.2	13.6	3.7		
	MCDB	-4.9	-1.4	-0.3	10.6	30.1	42.9	52.5	47.8	35.5	25.4	14.0	4.2			
	MDBR	9.1	9.0	9.1	8.5	6.9	6.8	9.5	7.7	5.3	6.9	8.5	8.9			
	5% DB	MCDBR	13.2	13.6	12.0	10.3	7.7	10.5	12.9	11.2	7.6	6.0	10.0	11.7		
(36) Mean Daily Temperature Range	MCWBR	16.7	13.0	11.4	9.5	6.1	7.1	6.8	6.5	5.4	5.1	8.8	8.2			
	5% WB	MCDBR	12.1	13.2	11.4	11.2	7.6	10.7	12.2	10.2	6.5	4.9	8.7	8.8		
	MCWBR	11.9	13.0	11.2	10.6	6.1	7.4	6.9	6.0	5.1	5.0	8.9	8.6			
	taub	0.150	0.196	0.235	0.282	0.310	0.312	0.322	0.309	0.268	0.214	0.148	N/A			
(40) Clear-Sky Solar Irradiance	taud	1.874	2.086	2.232	2.135	2.148	2.358	2.500	2.532	2.464	2.103	1.906	N/A			
	Ebn at Noon	32	199	261	273	277	281	275	267	250	179	27	0			
	Edh at Noon	2	13	23	34	38	32	27	23	19	13	2	0			
	RadAvg	9	162	682	1448	2005	1859	1550	1060	545	176	27	0			
(44) All-Sky Solar Radiation	RadStd	1	11	26	44	118	279	152	108	50	35	3	0			

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (2 neighbors)		+1.40	+2.50	N/A	N/A	N/A	N/A	-508	-506	N/A	N/A	

Nomenclature: See separate page